Week 12

Autoscaling and Cloud Automation for ML Workloads

- Application Phase
- Building practical cloud skills for machine learning operations
- Preparing for advanced GPU and distributed training weeks

Learning Objectives

- Understand cloud scaling principles and why they matter for ML workloads.
- Configure autoscaling for Azure virtual machines to match changes in job demand.
- Automate deployment and management tasks using Bash and Azure command-line tools.
- Monitor and control resource usage and costs with Azure monitoring tools.

This Week's Agenda

1. Scaling principles for cloud ML workloads
2. Autoscaling with Azure practical steps
3. Automating deployment with scripts
4. Monitoring resources and controlling costs
5. Hands-on activity and case example

Why Scale in the Cloud?

- Machine learning jobs can be highly variable in resource usage.
- Autoscaling helps match compute power to workload spikes; saves costs and increases efficiency.
- Industry demand for engineers who can optimize cloud operations for performance and spend.

Autoscaling in Azure for ML

- Azure supports virtual machine scale sets and automatic resource allocation.
- Autoscaling triggers can include CPU load, GPU utilization, or queue backlog.
- Example scenario; a company training NLP models scales up GPUs at night for batch jobs, then scales down for cost savings during idle times.

Automate Cloud Tasks Using Bash and CLI Tools

- Bash scripting enables rapid, repeatable automation of cloud setup and teardown processes.
- Azure CLI commands can start, stop, and scale VMs from scripts.
- Essential for high performance computing and production ML workflows.
- Example code to spin up a VM:

```
az vm create --resource-group ml-workshop --name gpu-vm01 --image UbuntuLTS --size Standard_NC6
```

- Scripts can be versioned for easy collaboration; aligns with industry best practices.

Monitoring and Cost Control in Azure

- Use Azure Monitor to track VM utilization, storage, and costs.
- Set alerts for abnormal spending or compute spikes to avoid overruns.
- Example; Configure dashboards to visualize GPU usage during ML training.
- Good monitoring is central to responsible and sustainable AI practice.

Activity. Automate a Cloud Workflow

- Write a Bash script that deploys or scales an Azure VM for ML training.
- Include steps for resource creation, monitoring setup, and cleanup.
- Test your script; document the process and common errors.

Industry Case Example

- Local startup needs to run nightly image classification jobs on large datasets.
- Engineers write scripts that detect job queue spikes and auto-scale GPU VMs for processing.
- Azure Monitor tools track costs in real-time for reporting to the business.
- Following cloud automation and monitoring best practices improves delivery reliability.

Linking To Previous Weeks

- Week 9. Securing database credentials with Key Vault scripts prepares for automated deployments.
- Week 10. Portfolio integration of Bash and Python skills underpin this week's scripting focus.
- Week 11. Monitoring model experiments ties into Azure resource monitoring and versioning.

Assessment Preparation. Next Steps

- Practice writing and testing automation scripts on Azure; document your workflows.
- Review Azure monitoring dashboards and experiment with different alert triggers.
- Next week; advanced GPU training and optimization workshops; continue building on your cloud automation foundations.
- All assessments require demonstration of automation, scaling, and monitoring skills.

Questions and Discussion

- Where would autoscaling make the biggest difference in your ML project plans?
- What challenges might arise in automating cloud infrastructure for ML workloads?
- How might you use monitoring to manage both performance and ethical AI concerns?